WORKSHEET

Predictions and residuals

1 The regression line:

$$\textit{Mark in test} = 7.8 + 12.3 \times \textit{Number of hours studied}$$

models the relationship between the mark obtained in a maths test and the number of hours studied, based on a group of students who studied between 0 and 5 hours for the test.

a Predict the mark for a student who studied 3 hours for the test. Is this prediction classified as interpolation or extrapolation?

b Predict the number of hours of study required to achieve a mark of 93.9? Is this prediction classified as interpolation or extrapolation?

c Which of the above predictions is the most reliable? Justify your answer.

d Use your answer to part **a** to calculate the residual for Aschtun who studied for 3 hours and got a mark of 57.

2 The least-squares regression line that determines a household heating bill for the winter by the average temperature is:

$$\textit{Heating bill (\$)} = 1428 - 56.9 \times \textit{Average outdoor temperature}$$

This relationship was determined by the heating bills of homes for which the average outdoor temperature was between 9 and 18 degrees.

a Predict the heating bill for a house where the average outdoor temperature was 5°. Is this prediction classified as interpolation or extrapolation?

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b Predict the outdoor temperature that would result in a heating bill of $574.50. Is this prediction classified as interpolation or extrapolation?

c Which of the above predictions is the most reliable? Justify your answer.

d Use your answer to part **b** to calculate the residual for a household in a climate where the average outdoor temperature was 15° and the heating bill was $498.85.

3 The regression line:

$$\textit{Muscle gain (kg)} = 0.013 \times \textit{Number of hours of exercise}$$

is based on data from a group of people who exercised between 20 and 70 hours.

a Predict the muscle gain for John, who exercised for 14 hours. Is this prediction classified as interpolation or extrapolation?

b Predict the number of hours of exercise required to gain 0.78 kg. Is this prediction classified as interpolation or extrapolation?

c Which of the above predictions is the most reliable? Justify your answer.

d Use your answer to part **b** to calculate the residual for Kim who exercised for 60 hours and gained 0.51 kg in muscle.

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4 The regression line:

$$\textit{Maximum sign legibility distance (m)} = 175.9 - 0.91 \times \textit{Age of driver}$$

is based on data from people aged between 25 and 58 years of age.

a Predict the maximum sign legibility distance for Susan, aged 39. Is this prediction classified as interpolation or extrapolation?

b Predict the age of a driver with a maximum sign legibility distance of 157.7 m. Is this prediction classified as interpolation or extrapolation?

c Which of the above predictions is the most reliable? Justify your answer.

d Use your answer to part **b** to calculate the residual for Miho, 20 years of age, with a maximum sign legibility distance of 118.4 m.

5 The hand spans and heights (both measured in cm) of a group of people were recorded. The heights of this group ranged from 158 cm to 177 cm. The regression model for the data was found to have the equation:

$$\textit{Hand span} = -3 + 0.14 \times \textit{Height}$$

a Predict the hand span for a person whose height is 164 cm. Is this prediction classified as interpolation or extrapolation?

b Use the equation to predict the height of Tony, whose hand span is 23.64 cm. Is this prediction classified as interpolation or extrapolation?

c Which of the above predictions is the most reliable? Justify your answer.

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d Use your answer to part **b** to calculate the residual for Tony, whose height is actually 186 cm.

6 The regression line:

Number of singles sold in a week = 18 096 − 389 × *Number of weeks the single has been in the Top 40*

is based on data for singles that had been in the Top 40 for between 10 and 26 weeks.

a Predict the number of singles sold in a week for a single in its 8th week in the Top 40. Is this prediction classified as interpolation or extrapolation?

b Use the equation of the regression line to estimate how many weeks a single has been in the Top 40 if its sales for the week are 9538. Is this prediction classified as interpolation or extrapolation?

c Which of the above predictions is the most reliable? Justify your answer.

d Use your answer to part **b** to calculate the residual for Justin B, whose sales this week are 15 768 for his single that has been in the Top 40 for 22 weeks.

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Answers

1 **a** 44.7; interpolation

b 7 h; extrapolation

c Prediction **a** is the most reliable because predictions classified as extrapolation tend to be less reliable than interpolation, as the linear regression line models the data given and may not represent values outside the data range as effectively.

d 12.3

2 **a** $1143.50; extrapolation

b 15°; interpolation

c Prediction **b** is the most reliable because predictions classified as extrapolation tend to be less reliable than interpolation, as the linear regression line models the data given and may not represent values outside the data range as effectively.

d −$75.65

3 **a** 0.182 kg; extrapolation

b 60 hours; interpolation

c Prediction **b** is the most reliable because predictions classified as extrapolation tend to be less reliable than interpolation, as the linear regression line models the data given and may not represent values outside the data range as effectively.

d −0.27 kg

4 **a** 140.41 m; interpolation

b 20 years old; extrapolation

c Prediction **a** is the most reliable because predictions classified as extrapolation tend to be less reliable than interpolation, as the linear regression line models the data given and may not represent values outside the data range as effectively.

d −39.3 m

5 **a** 19.96 cm; interpolation

b 190.3 cm; extrapolation

c Prediction **a** is the most reliable because predictions classified as extrapolation tend to be less reliable than interpolation, as the linear regression line models the data given and may not represent values outside the data range as effectively.

d −4.3 cm

6 **a** 14 984 singles; extrapolation

b 22 weeks; interpolation

c Prediction **b** is the most reliable because predictions classified as extrapolation tend to be less reliable than interpolation, as the linear regression line models the data given and may not represent values outside the data range as effectively.

d 6230 singles

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